

Supplementary Information for “Planar versus embedded PbS quantum-dot integration in perovskite solar cells”

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This document collects the supporting data referenced in the main text: the record of unsuccessful and excluded configurations, the FDTD numerical checks (mesh accuracy, polarization, incidence angle, effective-medium formulation), and the full electrical parameter sweeps. All efficiencies are referenced to 100 mW cm^{-2} and correspond to the unfiltered limit unless stated otherwise; J_{opt} denotes the integrated optical current and J_{sc} the short-circuit current returned by the drift-diffusion solver. Values reproduce from the named scripts and result files in the accompanying repository.

S1. UNSUCCESSFUL AND EXCLUDED CONFIGURATIONS

The following configurations were attempted and are excluded from the performance analysis, either because they exceeded the scope of the implemented numerical model or because they did not reach the solver. They are recorded here for completeness.

- *Tunnel-junction stack.* A tunnel-recombination junction between the quantum-dot and perovskite layers collapsed because the CHARGE model used carries no explicit band-to-band tunneling process. This is a limitation of the available transport model and is not read as evidence against the physical concept.
- *Minority-carrier-only interface recombination.* The interface boundary condition defaults to acting on minority carriers only. Scans configured this way produced no measurable change until both carrier channels were enabled; those early scans are excluded and all reported interface results use both carriers.
- *Blend effective-bandgap scan.* A one-at-a-time scan of the D2 blend effective bandgap over 1.40–1.50 eV returned an efficiency change of identically zero, inconsistent with the exponential dependence of the intrinsic carrier density, and hence of the open-circuit voltage, on bandgap. The scan is treated as not having reached the solver and is excluded from the sensitivity analysis; the influence of the assumed collection edge is instead quantified optically in the main text, through the change in the blend’s above-gap fraction when the edge is displaced.

- *Conformal grating in CHARGE.* A conformal (textured) grating cross-section was constructed but could not be driven with a correctly distributed generation profile across the widened simulation domain. All reported grating results therefore use the laterally averaged flat electrical geometry.
- *Optical mesh accuracy 4 for optimized cells.* Accuracy 4 could not be completed within the available compute budget for the widened grating unit cells; convergence for the optimized cells is reported at accuracy 3 (Table S1).

S2. FDTD MESH CONVERGENCE

Table S1 lists the integrated optical current against FDTD mesh accuracy. The optimized cells behind the headline efficiencies are converged: the optimized embedded (D2) cell changes by 1.3% between accuracy 2 and 3 (49.90 to 50.57 mA cm^{-2}), and the optimized planar (D1) grating cell by 1.7% (42.65 to 43.38 mA cm^{-2} , the latter from a separately converged run; accuracy 4 exceeded memory for the widened D1 cell). The baseline flat cells are reported at accuracy 2 as a common starting point; their larger accuracy 2-to-3 change does not affect the morphology comparison, which rests on the converged optimized and spectrally filtered results.

TABLE S1. FDTD mesh convergence of the integrated optical current.

Model	Mesh accuracy	J_{opt} (mA cm^{-2})
Baseline planar (D1)	2	39.13
	3	40.79
	4	40.96
Baseline embedded (D2)	2	42.42
	3	43.17
	4	44.24
Optimized planar (D1, grating)	2	42.65
	3	43.38
Optimized embedded (D2, grating)	2	49.90
	3	50.57

S3. POLARIZATION AND INCIDENCE ANGLE

The headline optical currents are TM-polarized. Table S2 gives the TE, TM and unpolarized (mean) currents for the optimized cells; the polarization asymmetry

is below 1% and the unpolarized efficiencies lie within 0.12 percentage points of the TM headline values. Table S3 gives the optical current versus angle of incidence for the flat and grating D1 cells; the response is weak over 0–30°.

TABLE S2. Polarization dependence of the optimized cells. Asymmetry is $|J_{\text{TM}} - J_{\text{TE}}|/J_{\text{unpol}}$.

Cell	J_{TM}	J_{TE}	J_{unpol}	Asym. (%)	PCE _{unpol} (%)
D1 optimized	42.81	42.52	42.63	0.7	23.89
D2 optimized	50.57	50.17	50.37	0.8	24.09

TABLE S3. Optical current versus angle of incidence (planar D1).

Configuration	θ (deg)	J_{opt} (mA cm ⁻²)
Flat	0	39.13
	15	39.29
	30	39.66
Grating	0	42.56
	30	41.44

S4. EFFECTIVE-MEDIUM FORMULATION

The blend dielectric function was computed with a Maxwell–Garnett mixing rule. As a cross-check, the complex refractive index of the 25 vol.% blend was re-computed with a Bruggeman effective medium over 300–1700 nm. The two formulations agree to a mean difference of ~1% and a maximum of 1.7% in the real index (largest near 400 nm), with a comparable difference in the extinction coefficient. The choice of mixing rule is therefore not a significant source of optical uncertainty. Both are optical approximations and neither asserts electrical equivalence between a homogeneous blend and discrete dots; the explicit square, hexagonal and honeycomb 5 nm dot patterns used to bound the optical error are described in the main text.

S5. QUANTUM-DOT MATERIAL SCREEN

Table S4 lists the six measured PbS datasets carried into the device model, spanning 0.80–1.24 eV and two ligand chemistries. The 1.03 eV NH₄I material gives the best current–voltage balance and is used for the planar comparison. Oleic-acid-capped dots were excluded before the electrical stage because the insulating ligand limits the mobility to order 10⁻⁵ cm² V⁻¹ s⁻¹.

TABLE S4. Quantum-dot bandgap and ligand screen (planar device).

Material	E_g (eV)	J_{sc}	V_{oc} (V)	FF	PCE (%)
NH ₄ I	1.033	38.49	0.737	0.719	20.38
OA	1.240	30.07	0.826	0.738	18.35
OA	1.127	28.59	0.770	0.752	16.55
NH ₄ I	0.905	36.71	0.658	0.675	16.31
OA	0.855	34.74	0.611	0.657	13.93
NH ₄ I	0.800	41.69	0.568	0.611	14.46

S6. EMBEDDED-DEVICE TRANSPORT AND DOPING

The embedded (D2) composite active layer is represented electrically by a homogeneous 1.44 eV surrogate (main-text Table II), and the optimized embedded device is sensitive to the transport values assigned to it. Increasing the blend carrier mobility from 0.02 to 50 cm² V⁻¹ s⁻¹ raises the fill factor from 0.576 to 0.678 and the unfiltered PCE from 21.94% to 24.30%, while V_{oc} falls from 0.827 to 0.714 V; a mobility of 20 cm² V⁻¹ s⁻¹ is used for the optimized comparison. The p-type doping produces a second voltage–current trade-off: at 10¹⁶ cm⁻³ the PCE reaches 24.91% (the high-voltage row of main-text Table I), and the highest sampled unfiltered point reaches 27.50% at 10¹⁷ cm⁻³. These mobility and high-doping values are upper-envelope sensitivities rather than evidence that a fabricated blend attains them, and the 27.50% point is excluded from the morphology comparison because it has no strict single-gap counterpart. Depth grading of the dot fraction did not improve D2: both tested grading directions reduced the efficiency below the 20.27% uniform reference, consistent with the embedded cell being absorption- rather than collection-limited in this model.

S7. INTERFACE RECOMBINATION

Table S5 gives the efficiency of both morphologies against a surface-recombination velocity S applied at the outer active interfaces. The two architectures degrade at markedly different rates: the planar device loses 6.0 points over the range while the embedded device loses 3.0, and the embedded cell overtakes the planar one already at $S = 10^2$ cm s⁻¹. This tolerance is the embedded morphology’s clearest electrical advantage, and it survives the spectral filtering that removes its apparent current benefit; the qualification, discussed in the main text, is that the sweep loads only the planar interfaces the model resolves and not the quantum-dot/perovskite interface distributed through the blend. Table S6 decomposes the planar loss by interface: the internal MAPbI₃/QD junction is the dominant channel, falling to 8.26% at 10⁶ cm s⁻¹, while the QD/EDT and SnO₂/MAPbI₃ interfaces are far less sensitive.

TABLE S5. Interface-recombination sweep with S applied at the outer active interfaces (unfiltered PCE, %). The embedded cell is the more tolerant of the two and overtakes the planar cell by $S = 10^2 \text{ cm s}^{-1}$.

$S \text{ (cm s}^{-1}\text{)}$	Planar (D1)	Embedded (D2)
0	20.59	20.27
10^2	20.01	20.26
10^3	19.55	20.23
10^4	18.34	19.91
10^5	15.79	18.47
10^6	14.58	17.23

TABLE S6. Per-interface recombination for the planar device (unfiltered PCE, %), one interface loaded at a time with both carrier channels enabled.

Interface	10^2	10^4	10^6 cm s^{-1}
QD/EDT	20.04	19.69	19.69
MAPbI ₃ /QD (internal)	17.18	11.10	8.26
SnO ₂ /MAPbI ₃	20.56	18.92	14.93

S8. GRATING AND LIGHT TRAPPING

Table S7 lists the flat-to-textured comparison for both morphologies. The reported figures are quoted at their stated grating configuration labels and are not merged with the separately optimized final devices. As noted in the main text, the grating enters the electrical model only through a laterally averaged generation profile.

TABLE S7. Rear-contact texturing (unfiltered). D1: 600 nm period, 200 nm tooth; D2: 600 nm period, 350 nm tooth.

Cell	Configuration	J_{sc}	PCE (%)
D1	Flat	40.03	21.14
	Grating	42.14	22.34
D2	Flat	41.45	19.88
	Grating	46.62	22.30

S9. DEPTH-RESOLVED SPECTRAL FILTERING

The strict single-gap results in the main text scale each absorber’s generation profile by a single AM1.5G-weighted above-gap fraction. This preserves the layer-integrated above-gap current exactly but not its depth distribution, since sub-gap photons penetrate differently from above-gap photons. As a check, the sub-gap generation was instead removed as a depth-uniform background—the weak-absorption limit for the sub-gap near-infrared—holding the retained above-gap current identical, and the optimized planar cell was re-solved (Table S8). Its strict single-gap PCE changes by only +0.11 points (18.80% to 18.91%). Because the filter acts identically on each absorber and every ac-

tive layer retains the great majority of its generation, the same depth-redistribution correction is small; the intra-absorber depth redistribution that distinguishes a wavelength-resolved filter from the scalar filter is therefore immaterial to the comparison.

TABLE S8. Strict single-gap PCE under the scalar above-gap filter and a depth-resolved (wavelength-aware) filter with identical retained current.

Cell	Scalar (%)	Depth-resolved (%)	Δ (pts)
D1 optimized	18.80	18.91	+0.11

S10. SENSITIVITY TO THE EMBEDDED ELECTRICAL REPRESENTATION

The embedded strict single-gap result depends on how the dispersed quantum dots are represented electrically, and this is the largest single sensitivity in the embedded comparison. The main text represents the blend as a homogeneous semiconductor with a volume-weighted 1.44 eV effective gap; at this edge the blend retains 59.1% of its integrated optical current under filtering and the optimized embedded cell reaches 14.23% (strict), behind the planar cell’s 18.80%. That edge is an averaging choice rather than a measured quantity, and a homogeneous surrogate cannot by construction distinguish a near-infrared photon absorbed above the gap of a discrete 1.03 eV dot from one absorbed below the 1.55 eV host gap.

Relaxing that averaging is equivalent to displacing the effective collection edge toward the 1.03 eV dot gap—the optimistic assumption that every dot-absorbed near-infrared photon is collected. The retained above-gap fraction of the blend absorptance then rises from 0.52 at the 1.55 eV host edge, through 0.59 at the 1.44 eV volume-weighted edge used in the main text, to 0.85 at the 1.03 eV dot edge (Table S9). Because the discarded current sits in exactly the near-infrared the embedded morphology harvests best, the embedded strict efficiency climbs steeply across this range and reaches parity with, then exceeds, the planar cell as the edge approaches the dot gap. The same effect seen electrically—representing the blend with its true 1.03 eV dot phase rather than a host-weighted average lifts the embedded ranking—is therefore the collection edge acting through the electrical model rather than an independent parameter.

This is why the main text does not assign the embedded architecture a single strict efficiency. The two collection limits bracket the outcome: the strict single-gap limit is the conservative endpoint, in which the dot near-infrared is filtered at the host-weighted 1.44 eV average and discarded, and the unfiltered limit the optimistic one, in which it is fully collected. Which morphology wins under strict filtering is decided by where the true collection edge falls, and a homogeneous single-gap model cannot fix it. Settling it requires a direct sub-bandgap

external-quantum-efficiency measurement on a blend at working loading, or the explicit-dot electrical model with a solved carrier-escape process—both identified as the decisive next steps in the main text.

TABLE S9. Above-gap fraction of the embedded-blend absorptance retained under strict filtering, versus the assumed effective collection edge, over the range bounded by the host (1.55 eV) and dot (1.03 eV) gaps. The 1.44 eV row is the volume-weighted value used in the main text; $\lambda_g = 1239.842/E_g$.

Assumed collection edge	λ_g (nm)	Retained above-gap fraction
1.55 eV (host gap)	800	0.52
1.44 eV (volume-weighted, main text)	861	0.59
1.03 eV (dot gap)	1204	0.85

S11. MATERIAL-PARAMETER PROVENANCE

Table S10 lists the source of each optical constant dataset and Table S11 the electrical parameters and their basis. All datasets and builder scripts are in the accompanying repository; the file paths below are relative to its root. Optical constants are experimental n, k tables except for the non-absorbing SiO_2 superstrate and the Au electrode, which use tabulated database values. Electrical parameters are grouped into consensus device-simulation values and the small number of best-case transport values used only for the optimized rows, which are flagged as upper-bound sensitivities in the main text.

TABLE S10. Optical constant (n, k) provenance.

Material	Source (repository <code>data/optical/</code>)
SiO_2 superstrate	Non-absorbing dielectric, database
FTO	Experimental n, k ; <code>layers/FTO_optical_constants.txt</code>
SnO_2	Experimental n, k ; <code>layers/SnO2_optical_constants.txt</code>
MAPbI_3	Experimental n, k ; <code>layers/perovskite.txt</code>
PbS-QD	Experimental n, k , size- and ligand-resolved; <code>qd/by_size, qd/b</code>
PbS-EDT	Experimental n, k ; <code>layers/PbS-EDT_optical_constants.txt</code>
MoO_x	Experimental n, k ; <code>layers/moo3.csv</code>
Au	CRC/Lumerical database

S12. CONTACT TREATMENT

Both electrical contacts are idealized as ohmic boundaries in CHARGE. The Au/ MoO_x anode and FTO cathode therefore enter the transport problem only through their assigned work functions, and because the boundary is forced ohmic those work functions are inactive: Table S12 shows that sweeping the anode work function over 5.1–5.7 eV and the cathode over 4.2–4.6 eV leaves J_{sc} , V_{oc} , FF and PCE unchanged for both morphologies. This is documented not as evidence that contact selectivity is unimportant but to make explicit that hole-selectivity, contact barriers and series contact resistance at the Au/ MoO_x interface lie outside the present model. The standalone MoO_3 transport material did not converge in CHARGE and is resolved optically only, folded electrically into the high-work-function anode; a barrier-resolved contact model is identified as future work in the main text.

TABLE S11. Electrical parameter provenance. Values in parentheses are the swept ranges. Best-case transport values used only for the optimized rows are marked $\dagger$ and are stated as upper-bound sensitivities in the main text.

Layer	E_g (eV)	χ (eV)	μ ($\text{cm}^2\text{V}^{-1}\text{s}^{-1}$)	Doping (cm^{-3})	SRH τ	Source / basis
SnO ₂	3.44	4.00	—	$n\,10^{17}$	—	ETL consensus
MAPbI ₃	1.55	3.90	10 (1–50)	$p\,10^{15}$ (to 10^{18})	$0.1\text{--}0.5\,\mu\text{s}^\dagger$	Gap/EA consensus; τ Stranks 2013, deQuilettes 2015
PbS-QD (D1)	1.03	3.85	0.02 (0.1 †)	$n\,10^{15}$	$10\,\mu\text{s}$	Gap measured (this work); μ Chuang 2014, Balazs 2017, Kagan 2017
D2 blend surrogate	1.44	3.92	10–20	$p\,10^{15}$ (to 10^{17})	$5\,\mu\text{s}$	Volume-weighted $0.20\times1.03+0.80\times1.55$; see main text
PbS-EDT	1.14	3.90	—	$p\,10^{16}$	—	p-type hole-contact layer

Contacts: FTO cathode 4.4 eV, Au/MoO_x anode 5.3 eV, both idealized ohmic (Sec. S10).

TABLE S12. Contact work-function sweep (forced-ohmic contacts). The response is flat by construction, confirming the contacts are idealized rather than barrier-resolved.

Design	Contact	Range (eV)	J_{sc}	V_{oc}	FF	PCE (%)
D1	anode	5.1–5.7	39.07	0.736	0.716	20.59
D1	cathode	4.2–4.6	39.07	0.736	0.716	20.59
D2	anode	5.1–5.7	42.58	0.740	0.709	22.32